

Mehmet KORUTURK

[in mehmek-koruturk210199](#) | [✉ mehmek-koruturk@vt.edu](mailto:mkoruturk@vt.edu) | [☎ +1 540 558 56 58](tel:+15405585658)

RESEARCH INTERESTS

My research focuses on reinforcement learning-driven decision-making for resilient and sustainable power systems. I develop model-free and model-based RL frameworks for coordinating distributed energy resources (e.g., PV, ESS, EV/FCEV systems) under uncertainty and non-stationary grid conditions. My work integrates stochastic and mixed-integer optimization with learning-based control to design adaptive energy management strategies that improve reliability, operational efficiency, and post-disaster resilience in grid-interactive systems.

EDUCATION

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|------------------------|--|
| August 2024 – Present | M.S. in Electrical Engineering
Virginia Polytechnic Institute and State University (Virginia Tech),
GPA: 3.95/4.00
Advisor: Ming Jin
Thesis (in progress): Benchmarking and Adaptive Reinforcement Learning for Sustainable and Resilient Power Systems
Relevant Coursework: <ul style="list-style-type: none">• Power System Planning (Prof. Lamine M. Mili)• Advanced Alternative Energy Systems (Prof. Saifur Rahman)• Advanced Microgrids (Prof. Ali Mehriizi-Sani)• Computational Methods in Power (Prof. Lamine M. Mili) |
| August 2017 – May 2022 | B.Sc. in Electrical Engineering
Yildiz Technical University (YTU), GPA: 3.21/4.00
Advisors: Prof. Ozan Erdinc , Prof. Ali Rifat Boynuegri |

RESEARCH EXPERIENCE

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| Graduate Research Assistant, Virginia Tech | 2024 – Present |
| <ul style="list-style-type: none">• Established benchmarking simulations for evaluating reinforcement learning algorithms in power system environments with operational constraints, uncertainty, and non-stationary dynamics.• Developed model-free and hybrid RL-based control strategies for distributed energy resource coordination in grid-interactive homes.• Formulated resilience-aware optimization models for post-disaster energy restoration using constrained vehicle-to-home resources.• Analyzed the stability and robustness of learning-based energy management policies under stochastic and time-varying system dynamics.• Built scalable simulation pipelines to assess algorithmic performance across large parameter spaces and uncertainty realizations. | |

PUBLICATIONS

Mehmet Koruturk, Alper Kagan Candan, et al. (July 2025). “Post-Disaster Energy Management for Homes Using Limited-Capacity Fuel Cell Electric Vehicle Batteries”. In: *Renewable Energy*. Under review.

Mehmet Koruturk, Artun Sel, et al. (June 2025). “Optimization of End-Consumer Energy Management with PV, ESS, EV, and Peer-to-Peer Trading in Smart Grids”. In: URL: <https://ieeexplore.ieee.org/document/11255815>.

Sel, Artun, Will Barrett, et al. (June 2025). “Estimation of Regions of Attraction for Nonlinear Systems via Coordinate-Transformed TS Models”. In: URL: <https://ieeexplore.ieee.org/document/11255827>.

Sel, Artun, Mehmet Koruturk, et al. (July 2025). *Estimation of Regions of Attraction for Nonlinear Systems via Coordinate-Transformed TS Models and Piecewise Quadratic Lyapunov Functions*. URL: <https://ieeexplore.ieee.org/document/11256030>.

Mehmet Koruturk, Bilgehan Sel, et al. (Apr. 2026). “Robust-Power Benchmark for Sustainable Power Systems”. In: *SmartGridComm*. Manuscript in preparation.

INDUSTRY EXPERIENCE

Electrical Engineer, Huawei Technologies Co. Ltd. Jul 2022 – Jul 2023, Istanbul, Turkey

- Led technical deployment and validation of 4G/5G base station installations.
- Conducted system-level acceptance testing and performance verification.
- Coordinated site implementation for large-scale telecom infrastructure projects.

PROFESSIONAL SERVICE

- Peer Reviewer,
IEEE Transactions on Automation Science and Engineering (T-ASE) (2025–Present)
- Core Organizing Committee Member,
Power and Energy Center (PEC) 2026 Annual Conference, Virginia Tech

AWARDS

- Awarded *Turkey’s Ministry of Education Scholarship for Graduate Studies in USA* (2022–Present)

SKILLS

Programming	Python, MATLAB
Engineering Tools	PSCAD, AutoCAD
Productivity	MS Office